Srijan Singh

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Summary

Final-year B.Tech Computer Science and Engineering undergraduate at SRMIST specializing in governed AI workflows, Generative AI, LLMs, prompt and context engineering, structured outputs, tool calling, and agentic multi-step pipelines. Experienced in Retrieval-Augmented Generation (RAG), vector databases, and evaluating LLM output quality through automated evals, test sets, and AI-led Quality Control (QC) layers. Proficient in Python, JavaScript, TypeScript, React, HTML, CSS, JSON data schemas, and Git/GitHub Actions to build reliable internal tools, versioned prompt repositories, and automated content and assessment validation pipelines.

Education

SRM Institute of Science and Technology, Kattankulathur — **Chennai, India**
B.Tech in Computer Science and Engineering | ***CGPA: 9.14/10*** — *2023–2027*

RPM Academy — **2022**
Class XII – Central Board of Secondary Education — *71.4%*

RPM Academy — **2020**
Class X – Central Board of Secondary Education — *78.8%*

Internship Experience

CodeAlpha — **Jul 2026 – Aug 2026**
AI Research & Workflow Engineering Intern | *Certificate* — *Remote*

- Built governed, multi-step AI workflows in Python using LLMs, prompt and context engineering, tool calling, and structured JSON outputs, enforcing phase sequencing and human review gates before final output generation.
- Implemented Retrieval-Augmented Generation (RAG) and vector database pipelines paired with an AI-led Quality Control (QC) layer to evaluate LLM output accuracy, completeness, and schema alignment.
- Designed automated evaluation suites (evals), benchmark test sets, and validation scripts to catch hallucinations and formatting errors early in the pipeline.

Prodigy Infotech — **Feb 2025 – Mar 2025**
Full-Stack AI Tooling & Schema Engineering Intern | *Certificate* — *Remote*

- Developed internal full-stack tools, REST APIs, and interactive authoring interfaces using Python, JavaScript, TypeScript, React, HTML, and CSS with standardized colours, card layouts, and UI components.
- Designed platform-ready JSON schemas and relational database structures to validate and store structured content data cleanly from the start.
- Migrated prompt files, context files, JSON schemas, and templates into versioned GitHub repositories with branching, pull requests, and automated validation via GitHub Actions.

Sukh Sagar Industries — **Mar 2025 – Apr 2025**
Quality Control & Assessment Pipeline Intern | *Certificate* — *Remote*

- Extended automated validation workflows to support structured question and assessment formats (MCQs, single-file and multi-file code submissions, and file uploads) with creation specs and rubric templates.
- Wrote and verified automated test cases to check code correctness, answer coverage, rubric validity, and specification alignment prior to deployment.
- Worked closely with content authors and internal teams to understand workflow pain points, feeding QC findings and review corrections back into source templates to continuously improve the pipeline.

Positions of Responsibility

IEEE SSIT, SRMIST
AI Workflow & Technical Lead

- Built internal web utilities and automated assessment-checking scripts using Python, TypeScript, React, and GitHub Actions for campus coding and AI workshops.
- Standardized JSON templates, problem creation specs, and test-case verification pipelines to ensure error-free programming challenges for hundreds of students.
- Collaborated with content creators and student mentors to gather feedback, document workflows, and ship rapid platform improvements.

Projects Undertaken

TRIPMACHA | Multi-Step Agentic AI Workflow, Tool Calling & QC Pipeline **Jun 2026**
Python / FastAPI / React / Google GenAI SDK / Structured JSON Outputs / Playwright / Docker GitHub
Domain: Governed AI Workflows | Prompt & Context Engineering | Tool Calling | Automated Evals

- Engineered a multi-step agentic AI workflow using Python, FastAPI, and Google GenAI SDK featuring strict phase sequencing, tool calling, and structured JSON output enforcement.
- Built an automated Quality Control (QC) and evaluation layer using Playwright and Python test sets to verify output accuracy, completeness, and schema compliance before rendering to users.
- Versioned prompt templates, context files, and JSON schemas in GitHub while standardizing interactive React card layouts and UI components.

AGRICOMPLY | RAG, Vector Database & AI-Led Quality Verification System **Jan 2026**
Python / Flask / React / RAG / Vector Database / JSON Schemas / MySQL / AWS S3 GitHub
Domain: RAG & Vector Databases | AI-Led QC Layer | Human Review Gates | Context Engineering

- Architected a Retrieval-Augmented Generation (RAG) and vector database pipeline in Python and React that grounds LLM responses against verified domain documents.
- Enforced system-level phase sequencing and human-in-the-loop review gates to validate document completeness, rubric criteria, and factual accuracy before final approval.
- Closed the feedback loop by logging review corrections and edge-case failures back into context files and prompt templates to continuously improve retrieval precision.

ReachInbox | Governed Phase-Sequenced Execution & Feedback Loop Engine **Oct 2026**
TypeScript / JavaScript / Next.js / Node.js / Express.js / BullMQ / Redis / PostgreSQL GitHub
Domain: Phase Sequencing Gates | Internal Full-Stack Tooling | Schema Validation | GitHub Workflows

- Developed a governed workflow orchestration engine in TypeScript, Node.js, and Next.js using BullMQ and Redis to enforce strict step-by-step execution and review gates.
- Designed platform-ready PostgreSQL and JSON data schemas to track workflow state transitions, reversible changes, and automated quality checks.
- Built a clean, standardized React dashboard for internal users to inspect pipeline outputs, resolve flagged errors, and feed corrections back into source templates.

DataLens | Platform-Ready Schema Validator, Assessment & Rubric Engine **Mar 2025**
TypeScript / JavaScript / React / Node.js / Express.js / MongoDB / HTML / CSS / Tailwind GitHub
Domain: JSON Schemas & Validation | Multi-Format File Uploads | Standardized UI/UX | Automated Checks

- Built a full-stack internal validation tool using TypeScript, React, and Node.js to ingest, parse, and verify multi-file uploads against strict JSON creation specs and rubric templates.
- Automated quality checks for data completeness, schema validity, and formatting consistency prior to publishing records to the platform database.
- Standardized UI colours, interactive card layouts, and visual diagnostics so authors and reviewers could spot and fix content errors effortlessly.

EmotionAI | Automated LLM/NLP Output Evaluation & Test-Set Benchmarking Suite **Jun 2026**
Python / TensorFlow / Keras / NLP / Automated Evals / PyTest / GitHub Actions GitHub
Domain: LLM & NLP Output Evals | Test Case Verification | Automated QC Checks | Benchmark Datasets

- Engineered an automated evaluation and benchmarking pipeline in Python to test NLP sequence models against curated gold-standard test sets, achieving 92.1% verified accuracy.
- Wrote deterministic PyTest test cases and quality checks to verify answer coverage, edge-case handling, and output consistency across model iterations.
- Integrated automated test execution into GitHub Actions so every prompt or model update was validated before merging.

Mentalytics | Structured Schema API & Automated Validation Pipeline **Feb 2026**
Python / FastAPI / JSON Schemas / Pydantic / Scikit-learn / Pandas / NumPy GitHub
Domain: Structured Outputs | API & Schema Design | Code Correctness | Documentation

- Built a Python and FastAPI service enforcing strict Pydantic JSON schemas, input validation, and structured outputs for downstream platform consumption.
- Wrote comprehensive unit test cases to verify algorithmic correctness, boundary conditions, and deterministic response formatting.
- Authored clear technical documentation and schema specifications to support seamless adoption by internal engineering and design teams.

ECOSHIELD | Interactive Learning & Visual Simulation Platform **Aug 2024**
Python | JavaScript | React | Flask | HTML | CSS | Tailwind CSS | REST APIs GitHub
Domain: Interactive UI Standards | Full-Stack Tooling | User-Centric Design | Rapid Iteration

- Designed and shipped an interactive full-stack web application using Python, Flask, React, and Tailwind CSS with standardized UI cards, consistent colour palettes, and interactive elements.
- Worked closely with end users to identify workflow friction points and rapidly ship UI and backend improvements from loosely defined problems to working solutions.
- Structured backend REST APIs and JSON payloads to ensure reliable, error-free rendering across interactive visual modules.

Certifications & Programs

SAP Certified | SAP Generative AI Developer **Mar 2026**
SAP Certificate
LLMs & Generative AI Workflows | Prompt & Context Engineering | Tool Calling & Structured Outputs | RAG

Deloitte Data Analytics Job Simulation **Sep 2026**
Deloitte | Forage Certificate
Quality Control (QC) Validation | Data Schemas & Integrity Checks | Structured Problem Solving | Documentation

Introduction to Machine Learning **Nov 2025**
NPTEL | IIT Kharagpur Certificate
Model Evaluation (Evals & Test Sets) | Automated Quality Checks | NLP & Deep Learning Foundations

Introduction to Database **Apr 2025**
NPTEL | IIT Madras Certificate
Platform-Ready Data Schemas | JSON & Relational Databases | Data Consistency | Query Optimization

Programming in Java **Oct 2024**
NPTEL | IIT Kharagpur Certificate
Code Correctness & Verification | Writing Test Cases | Single-File & Multi-File Programming Assessments

Networking Basics **Nov 2025**
Cisco Networking Academy Certificate
REST APIs & Web Protocols | Cloud Tooling Deployment | System Reliability

McKinsey.org Forward Program **2026**
Selected Participant | Core Skills Level Program Program
User Pain-Point Discovery | Extreme Ownership | Clear Written & Oral Communication | Iterative Improvement

Technical Skills

AI Workflows, GenAI & LLM Evaluation: Large Language Models (LLMs), Prompt Engineering, Context Engineering, Structured Outputs (JSON Schema / Pydantic), Tool Calling, Agentic & Multi-Step Workflows, Governed AI Workflows (Phase Sequencing & Human Review Gates), Retrieval-Augmented Generation (RAG), Vector Databases, Evaluating LLM Output Quality (Evals, Test Sets, Automated QC Checks), Google GenAI SDK

Programming, Full-Stack Tools & UI Standards: Python, JavaScript, TypeScript, React.js, Next.js, HTML5, CSS3, Tailwind CSS, FastAPI, Flask, Node.js, Express.js, RESTful APIs, Building Internal Full-Stack Tools & Scripts, Standardized UI/UX (Card Layouts, Colour Systems, Interactive Elements)

Version Control, Schemas & Assessment Pipelines: Git, GitHub (Versioned Prompt & Context Repositories, Branching, Pull Requests, Reversible Changes), GitHub Actions (CI/CD & Workflow Automation), JSON, Platform-Ready Data Schemas, PostgreSQL, MySQL, MongoDB, Redis, Writing & Verifying Test Cases, Code Correctness Verification (PyTest, Playwright), Assessment Pipelines (MCQ, Single-File & Multi-File Programming, File Upload, Rubric Templates)

Collaboration & Core Engineering Habits: Sitting with Authors & Product Teams to Solve Pain Points, Taking Loosely Defined Problems from Idea to Working Solution, Technical Documentation, Continuous Feedback Loop Integration